

BSSC RECRUITMENT CANDIDATE PORTAL

Codebase Architectural Reference & API Specifications

A comprehensive generated reference cataloging every source file, class definition, helper module, database schema, controller orchestrator, and utility function.

TOTAL FILES	TOTAL CLASSES	TOTAL FUNCTIONS
117	44	88

Generated automatically on June 30, 2026 | Platform Engine Version 1.0.0

ARCHITECTURAL OVERVIEW

This BSSC Candidate Portal Backend is built as an enterprise-grade Serverless Application utilizing TypeScript. It is designed to run efficiently on AWS Lambda using API Gateway triggers, maintaining a clean architectural decoupling of concerns. Below is a summary of the layout and flow of logic:

- **Controller Layer:** Encapsulates entry routing logic. Controllers receive events, invoke schema validations, execute business operations via services, and return normalized responses.
- **Service Layer:** Contains the core business rules and manages workflows (e.g. registration steps, payment status inquiries, document validations). Handles external integrations such as Cognito, Razorpay, GetEPay, AWS SES, and SMS gateways.
- **Data Access Layer (Repositories):** Executes database operations. Abstracts query creation using the Drizzle ORM and interfaces directly with the PostgreSQL connection pool.
- **Validation Layer (Validators):** Enforces request and response constraints at the system boundaries using Zod validation schemas.
- **Database Layer:** Definitively defines database tables, primary/foreign relationships, lookup seeders, and tracks SQL alterations via drizzle-kit migrations.

CODEBASE METRICS



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Orchestrators that receive API request payloads, invoke validators, and route actions to business services.

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[category.controller.ts](#) Contains class CategoryController

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[subject.controller.ts](#) Contains class SubjectController

src/database

Database ORM schemas, migration setup, seed configurations, and relations defined using Drizzle ORM.

[drizzle.ts](#) Defines functions getPool, getDb, closeDb

[dropOldTables.ts](#) Defines functions dropOldTables

[migrate.ts](#) Defines functions runMigrations

[relations.ts](#) Exports variables rolesRelations, usersRelations, candidatesRelations...

[schema.ts](#) Exports variables roles, users, candidates...

[truncate.ts](#) Defines functions truncateAllTables

src/database/seeders

Database seeder scripts that populate initial mock or lookup data.

[categories.seeder.ts](#) Defines functions seedCategories, seedRecursive

[degrees.seeder.ts](#) Defines functions seedDegrees

[disabilities.seeder.ts](#) Defines functions seedDisabilities

[index.ts](#) Defines functions runSeeders

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[posts.seeder.ts](#) Defines functions seedPostsRelatedData

[roles.seeder.ts](#) Defines functions seedRoles

[subjects.seeder.ts](#) Defines functions seedSubjects

[users.seeder.ts](#) Defines functions seedUsers

src/errors

Custom application error classes providing standard structure for error reporting.

[AppError.ts](#) Contains class AppError, ValidationError, UnauthorizedError, ForbiddenError, N...

src/handlers

AWS Lambda Entrypoints and Lambda handlers (Cognito events, custom SMS, background actions).

application.ts	Exports variables getDraft, getApplication, getAllStepsData...
category.ts	Exports variables seedCategories, createCategory, listCategories
customSMSSender.ts	Defines functions sendSms, handleCustomSMSSender, handler
dashboard.ts	Exports variables getDashboard, getNotifications, getAdmitCard...
degree.ts	Exports variables seedDegrees, createDegree, listDegrees
disability.ts	Exports variables listDisabilities
document.ts	Exports variables upload, getAll, getDownloadUrl...
gatewayPayment.ts	Exports variables create, verify, getStatus...
location.ts	Exports variables getCountries, getStates, getDistricts
payment.ts	Exports variables initiate, verify, getStatus...
post.ts	Exports variables listPosts
public.ts	Exports variables getAgeLimits, getCognitoSubld
subject.ts	Exports variables seedSubjects, createSubject, listSubjects
submission.ts	Exports variables submit

src/handlers/admin

AWS Lambda handlers for admin actions (data export, validation, statistics).

exportCandidatesXlsx.ts	Exports variables handler
getCandidateDetail.ts	Exports variables handler
getCandidateDocuments.ts	Exports variables handler
getStats.ts	Exports variables handler
listCandidates.ts	Exports variables handler
verifyDocument.ts	Exports variables handler

src/handlers/auth

AWS Lambda handlers specific to user, admin, and candidate authentication flow.

candidateRegister.ts	Exports variables handler
candidateStep1.ts	Exports variables handler
candidateStep2.ts	Exports variables handler
candidateStep3.ts	Exports variables handler
candidateStep4.ts	Exports variables handler
candidateStep5.ts	Exports variables handler
candidateStep6.ts	Exports variables handler
changePassword.ts	Exports variables handler
customMessage.ts	Defines functions handler, getOtpTemplate, getloginPasswordTemplate...
customSmsSender.ts	Defines functions handler
forgotPassword.ts	Exports variables handler
getAllCandidates.ts	Exports variables handler
getCandidateRegistration.ts	Exports variables handler
getCandidateStep.ts	Exports variables handler
getCaptcha.ts	Exports variables handler
getProfile.ts	Exports variables handler
login.ts	Exports variables handler
logout.ts	Exports variables handler
refreshToken.ts	Exports variables handler
register.ts	Exports variables handler
resetPassword.ts	Exports variables handler
seedDummy.ts	Exports variables handler
validateCaptcha.ts	Exports variables handler

src/handlers/notification

Submodule containing codebase files.

sendEmail.ts	Exports variables handler
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src/handlers/otp

AWS Lambda handlers for OTP creation, sending, and verification processes.

resendOtp.ts	Exports variables handler
sendEmailOtp.ts	Exports variables handler
sendMobileOtp.ts	Exports variables handler
verifyEmailOtp.ts	Exports variables handler
verifyMobileOtp.ts	Exports variables handler

src/helpers

Reusable utility functions for working with multipart payloads, HTTP requests, and standardized responses.

multipart.ts	Defines functions parseMultipart
request.ts	Defines functions parseBody, parseEvent, getCookieValue...
response.ts	Defines functions getOrigin, buildHeaders

src/middleware

HTTP intermediate logic processing (e.g., authentication verify, schema validations, error catching).

auth.ts	Defines functions authenticate, requireRole
errorHandler.ts	Defines functions errorHandler
validate.ts	Defines functions formatZodErrors, validate

src/repositories

Data Access Layer encapsulating direct database operations via Drizzle ORM.

admin.repository.ts	Contains class AdminRepository
application.repository.ts	Contains class ApplicationRepository
common.repository.ts	Contains class OtpRepository, CaptchaRepository, DocumentRepository, Paym...
gatewayPayment.repository.ts	Contains class GatewayPaymentRepository
location.repository.ts	Contains class LocationRepository
user.repository.ts	Contains class UserRepository

src/scripts

Submodule containing codebase files.

triggerMtsSms.ts	Defines functions triggerSms
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src/services

Business Logic Layer coordinating transactional workflows, third-party adapters (Razorpay, GetEPay, AWS SES/SNS).

admin.service.ts	Contains class AdminService
application.service.ts	Contains class ApplicationService
auth.service.ts	Contains class AuthService
document.service.ts	Contains class DocumentService
gatewayPayment.service.ts	Contains class GatewayPaymentService
getepay.adapter.ts	Contains class GetepayAdapter
notification.service.ts	Contains class NotificationService
otp.service.ts	Contains class OtpService
payment.service.ts	Contains class PaymentService
razorpay.adapter.ts	Contains class RazorpayAdapter

src/types

Common TypeScript definitions, models, and interface definitions.

[index.ts](#) =====

src/utls

General utilities including age calculation, captcha generation, AWS Cognito helper, email sending, SMS integration, and PDF helper.

age.ts	Safely parses a date string (e.g., DD-MM-YYYY, YYYY-MM-DD, DD/MM/YYYY)...
captcha.ts	Defines functions generateCaptcha, verifyCaptchaText
cognito.ts	Defines functions computeSecretHash, generateMockTokens, cognitoLogin...
crypto.ts	Defines functions generateHash, compareHash, generateOTP...
email.ts	Defines functions sendEmail, sendPaymentSuccessEmail
pdf.ts	Defines functions generateApplicationHtml
sms.ts	Defines functions sendSms, sendMtsExamNotificationSms, sendMtsExamPorta...

src/validators

Zod verification schemas for request body and query parameter validation.

application.ts	Exports variables step0Schema, step1Schema, step2Schema...
auth.ts	Exports variables loginSchema, validateCaptchaSchema, registerSchema...
gatewayPayment.ts	Exports variables createPaymentSchema, verifyPaymentSchema, refundPaym...
notification.ts	Exports variables sendEmailNotificationSchema

SRC/CONFIG

System configurations and environment variables loading.

src/config/index.ts

CONFIG

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local parseEnv()	No doc comment Params: none Returns: AppConfig

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
config	AppConfig	No description

SRC/CONTROLLERS

Orchestrators that receive API request payloads, invoke validators, and route actions to business services.

src/controllers/admin.controller.ts

CONTROLLERS

No description provided in file comments.

class AdminController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getStats(event)	— GET /admin/stats — Params: event: APIGatewayProxyEventV2 Returns: Promise
public listCandidates(event)	— GET /admin/candidates — Query params: page, limit, search, status, isSubmitted, mobileVerified, emailVerified, fromDate, toDate, sortBy, sortOrder Params: event: APIGatewayProxyEventV2 Returns: Promise
public getCandidateDetail(event)	— GET /admin/candidates/{candidateId} — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getCandidateDocuments(event)	— GET /admin/candidates/{candidateId}/documents — Query params: page, limit Params: event: APIGatewayProxyEventV2 Returns: Promise
public exportCandidatesXlsx(event)	— GET /admin/candidates/export/xlsx — Accepts same query filters as listCandidates (except page/limit) Returns base64-encoded XLSX Params: event: APIGatewayProxyEventV2 Returns: Promise
public verifyDocument(event)	— PATCH /admin/documents/{documentId}/verify — Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
adminController	any	No description

src/controllers/application.controller.ts

CONTROLLERS

No description provided in file comments.

class ApplicationController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getDraft (event)	— GET /application — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getApplication (event)	— GET /application/{applicationId} — Params: event: APIGatewayProxyEventV2 Returns: Promise
public saveStep (event)	— POST /application/{applicationId}/step/{stepNumber} — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getStep (event)	— GET /application/{applicationId}/step/{stepNumber} — Params: event: APIGatewayProxyEventV2 Returns: Promise
public submit (event)	— POST /application/{applicationId}/submit — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getPrintPreview (event)	— GET /application/{applicationId}/print — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getAllStepsData (event)	— GET /application/{applicationId}/steps or GET /application/steps/all — Params: event: APIGatewayProxyEventV2 Returns: Promise
public finalSubmitLegacy (event)	— POST /application/{applicationId}/submit-final (LEGACY) — Params: event: APIGatewayProxyEventV2 Returns: Promise
public unifiedFinalSubmit (event)	— POST /application/{applicationId}/final-submit — Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
applicationController	any	No description

src/controllers/auth.controller.ts

CONTROLLERS

No description provided in file comments.

class AuthController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getCaptha(_event)	— GET /auth/captha — Params: _event: APIGatewayProxyEventV2 Returns: Promise
public validateCaptha(event)	— POST /auth/captha/validate — Params: event: APIGatewayProxyEventV2 Returns: Promise
public login(event)	— POST /auth/login — Params: event: APIGatewayProxyEventV2 Returns: Promise
public register(event)	— POST /auth/register — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateRegister(event)	— POST /auth/candidate/register — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep1(event)	— POST /auth/candidate/step-1 — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep2(event)	— PATCH /auth/candidate/step-2 — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep3(event)	<i>No doc comment</i> Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep4(event)	— PATCH /auth/candidate/step-4 — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep5(event)	— PATCH /auth/candidate/step-5 — Params: event: APIGatewayProxyEventV2 Returns: Promise
public candidateStep6(event)	— PATCH /auth/candidate/step-6 — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getCandidateStep(event)	— GET /auth/candidate/step/{stepNumber} — Params: event: APIGatewayProxyEventV2 Returns: Promise
public refreshToken(event)	— POST /auth/refresh-token — Params: event: APIGatewayProxyEventV2 Returns: Promise

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public forgotPassword(event)	— POST /auth/forgot-password — Params: event: APIGatewayProxyEventV2 Returns: Promise
public resetPassword(event)	— POST /auth/reset-password — Params: event: APIGatewayProxyEventV2 Returns: Promise
public changePassword(event)	— POST /auth/change-password — Params: event: APIGatewayProxyEventV2 Returns: Promise
public logout(event)	— POST /auth/logout — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getProfile(event)	— GET /auth/profile — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getCandidateRegistration(event)	— GET /auth/candidate/registration — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getAllCandidates(event)	— GET /auth/candidates — Params: event: APIGatewayProxyEventV2 Returns: Promise
public seedDummy(_event)	— POST /auth/seed-dummy — Params: _event: APIGatewayProxyEventV2 Returns: Promise
public sendMobile0tp(event)	— OTP Endpoints — Params: event: APIGatewayProxyEventV2 Returns: Promise
public verifyMobile0tp(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public sendEmail0tp(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public verifyEmail0tp(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public resend0tp(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>evaluateAgeLimits</code> (age, categoryInput, categoryName, gender, isPwD)	<i>No doc comment</i> Params: age: number, categoryInput: string, categoryName: string, gender: string, isPwD: boolean Returns: boolean
local <code>resolveJobQualificationId</code> (level, degreeOrCourse, specializationOrSubject, providedId?)	<i>No doc comment</i> Params: level: string, degreeOrCourse: string, specializationOrSubject: string, providedId?: number Returns: number undefined

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
authController	any	<i>No description</i>

src/controllers/category.controller.ts

CONTROLLERS

No description provided in file comments.

class `CategoryController`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>seed</code> (_event)	— POST /categories/seed ————— Seeds all predefined categories & sub-categories Params: _event: APIGatewayProxyEventV2 Returns: Promise
public <code>create</code> (event)	— POST /categories ————— Insert a single category or sub-category via API Params: event: APIGatewayProxyEventV2 Returns: Promise
public <code>list</code> (_event)	— GET /categories ————— List all categories with their sub-categories nested Params: _event: APIGatewayProxyEventV2 Returns: Promise

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>buildTree</code> (parentId)	<i>No doc comment</i> Params: parentId: number null Returns: any[]

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>categoryController</code>	any	No description

src/controllers/dashboard.controller.ts

CONTROLLERS

No description provided in file comments.

class DashboardController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>getDashboard(event)</code>	— GET /dashboard — Params: <code>event: APIGatewayProxyEventV2</code> Returns: <code>Promise</code>
public <code>getNotifications(event)</code>	— GET /dashboard/notifications — Params: <code>event: APIGatewayProxyEventV2</code> Returns: <code>Promise</code>
public <code>getAdmitCard(event)</code>	— GET /dashboard/admit-card — Params: <code>event: APIGatewayProxyEventV2</code> Returns: <code>Promise</code>
public <code>getResult(event)</code>	— GET /dashboard/result — Params: <code>event: APIGatewayProxyEventV2</code> Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>dashboardController</code>	any	No description

src/controllers/degree.controller.ts

CONTROLLERS

No description provided in file comments.

class DegreeController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>seed(_event)</code>	— POST /degrees/seed — Seeds predefined degrees into the database Params: <code>_event: APIGatewayProxyEventV2</code> Returns: <code>Promise</code>

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public create (event)	— POST /degrees — Insert a single degree via API Params: event: APIGatewayProxyEventV2 Returns: Promise
public list (event)	— GET /degrees — List all degrees Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
degreeController	any	No description

src/controllers/disability.controller.ts

CONTROLLERS

No description provided in file comments.

class DisabilityController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public list (_event)	— GET /disabilities — List all disabilities Params: _event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
disabilityController	any	No description

src/controllers/document.controller.ts

CONTROLLERS

No description provided in file comments.

class DocumentController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public upload (event)	— POST /documents/upload — Params: event: APIGatewayProxyEventV2 Returns: Promise

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getAll (event)	— GET /documents — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getDownloadUrl (event)	— GET /documents/{documentId}/download — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getPresignedUrl (event)	— GET /documents/{documentId}/presigned-url — Params: event: APIGatewayProxyEventV2 Returns: Promise
public delete (event)	— DELETE /documents/{documentId} — Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
documentController	any	No description

src/controllers/gatewayPayment.controller.ts

CONTROLLERS

No description provided in file comments.

class GatewayPaymentController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public create (event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public verify (event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public getStatus (event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public refund (event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public webhook (event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public returnRedirect (event)	<i>No doc comment</i> Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
gatewayPaymentController	any	<i>No description</i>

src/controllers/location.controller.ts

CONTROLLERS

No description provided in file comments.

class LocationController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getCountries (_event)	— GET /countries — Params: _event: APIGatewayProxyEventV2 Returns: Promise
public getStates (event)	— GET /countries/{countryId}/states — Params: event: APIGatewayProxyEventV2 Returns: Promise
public getDistricts (event)	— GET /states/{stateId}/districts — Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
locationController	any	<i>No description</i>

src/controllers/notification.controller.ts

CONTROLLERS

No description provided in file comments.

class NotificationController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>sendEmail(event)</code>	<i>No doc comment</i> Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>notificationController</code>	any	<i>No description</i>

src/controllers/payment.controller.ts

CONTROLLERS

No description provided in file comments.

class PaymentController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>initiate(event)</code>	— POST /payment/initiate — Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>
public <code>verify(event)</code>	— POST /payment/verify — Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>
public <code>getStatus(event)</code>	— GET /payment/{applicationId}/status — Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>
public <code>getFeeStructure(_event)</code>	— GET /payment/fee-structure — Params: <code>_event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>
public <code>getInvoice(event)</code>	— GET /payment/{paymentOrderId}/invoice — Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>
public <code>freeReturn(event)</code>	— GET /payment/free/return — Params: <code>event</code>: APIGatewayProxyEventV2 Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>paymentController</code>	any	<i>No description</i>

src/controllers/post.controller.ts

CONTROLLERS

No description provided in file comments.

class PostController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public list(event)	— GET /posts — List all posts Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
postController	any	No description

src/controllers/public.controller.ts

CONTROLLERS

No description provided in file comments.

class PublicController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getAgeLimits(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
public getCognitoSubId(event)	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
publicController	any	No description

src/controllers/subject.controller.ts

CONTROLLERS

No description provided in file comments.

class SubjectController

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public seed (_event)	— POST /subjects/seed — Seeds all 15 predefined subjects into the database Params: _event: APIGatewayProxyEventV2 Returns: Promise
public create (event)	— POST /subjects — Insert a single subject via API Params: event: APIGatewayProxyEventV2 Returns: Promise
public list (_event)	— GET /subjects — List all subjects Params: _event: APIGatewayProxyEventV2 Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
subjectController	any	<i>No description</i>

SRC/DATABASE

Database ORM schemas, migration setup, seed configurations, and relations defined using Drizzle ORM.

src/database/drizzle.ts

DATABASE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>getPool()</code>	<i>No doc comment</i> Params: none Returns: Pool
export <code>getDb()</code>	<i>No doc comment</i> Params: none Returns: NodePgDatabase
export <code>closeDb()</code>	<i>No doc comment</i> Params: none Returns: Promise

src/database/dropOldTables.ts

DATABASE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>dropOldTables()</code>	<i>No doc comment</i> Params: none Returns: Promise

src/database/migrate.ts

DATABASE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>local runMigrations()</code>	No doc comment Params: none Returns: Promise

src/database/relations.ts

DATABASE

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>rolesRelations</code>	any	No description
<code>usersRelations</code>	any	No description
<code>candidatesRelations</code>	any	No description
<code>applicationsRelations</code>	any	No description
<code>applicationStepDataRelations</code>	any	No description
<code>documentsRelations</code>	any	No description
<code>paymentsRelations</code>	any	No description
<code>invoicesRelations</code>	any	No description
<code>categoriesRelations</code>	any	No description
<code>educationsRelations</code>	any	No description
<code>subjectsRelations</code>	any	No description
<code>postsRelations</code>	any	No description
<code>countriesRelations</code>	any	No description
<code>statesRelations</code>	any	No description
<code>districtsRelations</code>	any	No description

src/database/schema.ts

DATABASE

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
roles	any	— Roles
users	any	— Users
candidates	any	— Candidates
applications	any	— Applications
applicationStepData	any	— Application Step Data
finalSubmissions	any	— Final Submissions
documents	any	— Documents
payments	any	— Payments
invoices	any	— Invoices
otps	any	— OTPs
captchas	any	— CAPTCHAs
categories	any	— Categories
educations	any	— Educations
subjects	any	— Subjects
posts	any	— Posts
countries	any	— Countries
states	any	— States
districts	any	— Districts
jobQualifications	any	— Job Qualifications
candidateQualifications	any	— Candidate Qualifications
candidatePostPreferences	any	— Candidate Post Preferences

VARIABLE NAME	TYPE	DESCRIPTION
candidateLanguages	any	— Candidate Languages
degrees	any	— Degrees
degreePostMap	any	No description
postMapping	any	post_mapping
jobEligibility	any	No description
disabilities	any	— Exported — Disabilities
paymentGateways	any	— Payment Gateways
paymentTransactions	any	— Payment Transactions
paymentLogs	any	— Payment Logs
paymentRefunds	any	— Payment Refunds

src/database/truncate.ts

DATABASE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local truncateAllTables()	No doc comment Params: none Returns: Promise

SRC/DATABASE/SEEDERS

Database seeder scripts that populate initial mock or lookup data.

src/database/seeders/categories.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export seedCategories(userId)</code>	No doc comment Params: <code>userId: number</code> Returns: <code>Promise<{ inserted: string[]; skipped: string[] }></code>
<code>local</code> <code>seedRecursive(categoriesData, parentId, level)</code>	No doc comment Params: <code>categoriesData: any[], parentId: number null, level: number</code> Returns: <code>any</code>

src/database/seeders/degrees.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export seedDegrees()</code>	No doc comment Params: <code>none</code> Returns: <code>Promise<{ inserted: string[]; skipped: string[] }></code>

src/database/seeders/disabilities.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export seedDisabilities()</code>	No doc comment Params: <code>none</code> Returns: <code>Promise</code>

src/database/seeders/index.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local runSeeders()	No doc comment Params: none Returns: Promise

src/database/seeders/jobQualifications.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export seedJobQualifications()	No doc comment Params: none Returns: Promise

src/database/seeders/location.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export seedLocations()	No doc comment Params: none Returns: Promise

src/database/seeders/posts.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>seedPostsRelatedData()</code>	<i>No doc comment</i> Params: none Returns: <code>Promise</code>

src/database/seeder/roles.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>seedRoles()</code>	<i>No doc comment</i> Params: none Returns: <code>Promise</code>

src/database/seeder/subjects.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>seedSubjects(userId)</code>	<i>No doc comment</i> Params: <code>userId: number</code> Returns: <code>Promise<{ inserted: string[]; skipped: string[] }></code>

src/database/seeder/users.seeder.ts

SEEDERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>seedUsers()</code>	<i>No doc comment</i> Params: none Returns: <code>Promise</code>

SRC/ERRORS

Custom application error classes providing standard structure for error reporting.

src/errors/AppError.ts

ERRORS

No description provided in file comments.

class AppError

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public constructor (message, statusCode, errors)	<i>No doc comment</i> Params: message: string, statusCode: number, errors: Array<{ field?: string; message: string }> Returns: any

class ValidationError

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public constructor (errors, message)	<i>No doc comment</i> Params: errors: Array<{ field: string; message: string }>, message: string Returns: any

class UnauthorizedError

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public constructor (message)	<i>No doc comment</i> Params: message: string Returns: any

class ForbiddenError

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public constructor (message)	<i>No doc comment</i> Params: message: string Returns: any

class NotFoundError

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public constructor (message)	<i>No doc comment</i> Params: message: string Returns: any

class `ConflictError`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public constructor(message)</code>	<i>No doc comment</i> Params: <code>message: string</code> Returns: <code>any</code>

class `DatabaseError`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public constructor(message, internalError)</code>	<i>No doc comment</i> Params: <code>message: string</code> , <code>internalError: Error null</code> Returns: <code>any</code>

class `TooManyRequestsError`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public constructor(message)</code>	<i>No doc comment</i> Params: <code>message: string</code> Returns: <code>any</code>

class `UnsupportedMediaTypeError`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public constructor(message)</code>	<i>No doc comment</i> Params: <code>message: string</code> Returns: <code>any</code>

SRC/HANDLERS

AWS Lambda Entrypoints and Lambda handlers (Cognito events, custom SMS, background actions).

src/handlers/application.ts

HANDLERS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
getDraft	any	No description
getApplication	any	No description
getAllStepsData	any	No description
saveStep	any	No description
getStep	any	No description
getPrintPreview	any	No description
finalSubmitLegacy	any	No description
unifiedFinalSubmit	any	No description

src/handlers/category.ts

HANDLERS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
seedCategories	any	No description
createCategory	any	No description
listCategories	any	No description

src/handlers/customSMSSender.ts

HANDLERS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>sendSms(undefined)</code>	<i>No doc comment</i> Params: <code>undefined: SendSmsParams</code> Returns: <code>Promise</code>
local <code>handleCustomSMSSender(event)</code>	<i>No doc comment</i> Params: <code>event: CognitoEvent</code> Returns: <code>Promise</code>
export <code>handler(event)</code>	<i>No doc comment</i> Params: <code>event: CognitoEvent</code> Returns: <code>Promise</code>

src/handlers/dashboard.ts

HANDLERS

*No description provided in file comments.***Exported Variables & Schemas**

VARIABLE NAME	TYPE	DESCRIPTION
getDashboard	any	<i>No description</i>
getNotifications	any	<i>No description</i>
getAdmitCard	any	<i>No description</i>
getResult	any	<i>No description</i>

src/handlers/degree.ts

HANDLERS

*No description provided in file comments.***Exported Variables & Schemas**

VARIABLE NAME	TYPE	DESCRIPTION
seedDegrees	any	<i>No description</i>
createDegree	any	<i>No description</i>
listDegrees	any	<i>No description</i>

src/handlers/disability.ts

HANDLERS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>listDisabilities</code>	any	No description

src/handlers/document.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>upload</code>	any	No description
<code>getAll</code>	any	No description
<code>getDownloadUrl</code>	any	No description
<code>getPresignedUrl</code>	any	No description
<code>deleteDocument</code>	any	No description

src/handlers/gatewayPayment.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>create</code>	any	No description
<code>verify</code>	any	No description
<code>getStatus</code>	any	No description
<code>refund</code>	any	No description
<code>webhook</code>	any	No description
<code>returnRedirect</code>	any	No description

src/handlers/location.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
getCountries	any	No description
getStates	any	No description
getDistricts	any	No description

src/handlers/payment.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
initiate	any	No description
verify	any	No description
getStatus	any	No description
getFeeStructure	any	No description
getInvoice	any	No description
freePaymentReturn	any	No description

src/handlers/post.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
listPosts	any	No description

src/handlers/public.ts**HANDLERS**

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
getAgeLimits	any	No description
getCognitoSubId	any	No description

src/handlers/subject.ts

HANDLERS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
seedSubjects	any	No description
createSubject	any	No description
listSubjects	any	No description

src/handlers/submission.ts

HANDLERS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
submit	any	No description

SRC/HANDLERS/ADMIN

AWS Lambda handlers for admin actions (data export, validation, statistics).

src/handlers/admin/exportCandidatesXlsx.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/admin/getCandidateDetail.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/admin/getCandidateDocuments.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/admin/getStats.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/admin/listCandidates.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/admin/verifyDocument.ts

ADMIN

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

SRC/HANDLERS/AUTH

AWS Lambda handlers specific to user, admin, and candidate authentication flow.

src/handlers/auth/candidateRegister.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/candidateStep1.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/candidateStep2.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/candidateStep3.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
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VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/candidateStep4.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/candidateStep5.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	APIGatewayProxyHandlerV2	No description

src/handlers/auth/candidateStep6.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/changePassword.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/customMessage.ts

AUTH

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export handler (event, _context)	No doc comment Params: event: CustomMessageTriggerEvent, _context: Context Returns: Promise
local getOtpTemplate (name, code, message)	No doc comment Params: name: string, code: string, message: string Returns: string
local getloginPasswordTemplate (name, email, temporaryPassword?)	No doc comment Params: name: string, email: string, temporaryPassword?: string Returns: string
local getForgotPasswordTemplate (name, code)	No doc comment Params: name: string, code: string Returns: string

src/handlers/auth/customSmsSender.ts

AUTH

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export handler (event, _context)	No doc comment Params: event: CustomSMSSenderTriggerEvent, _context: Context Returns: Promise

src/handlers/auth/forgotPassword.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/getAllCandidates.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/getCandidateRegistration.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/getCandidateStep.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/getCaptcha.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/getProfile.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/login.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/logout.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/refreshToken.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/register.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/resetPassword.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/seedDummy.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/auth/validateCaptcha.ts

AUTH

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

SRC/HANDLERS/NOTIFICATION

Submodule containing codebase files.

src/handlers/notification/sendEmail.ts

NOTIFICATION

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

SRC/HANDLERS/OTP

AWS Lambda handlers for OTP creation, sending, and verification processes.

src/handlers/otp/resend0tp.ts

OTP

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/otp/sendEmail0tp.ts

OTP

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/otp/sendMobile0tp.ts

OTP

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/otp/verifyEmail0tp.ts

OTP

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

src/handlers/otp/verifyMobile0tp.ts

OTP

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
handler	any	No description

SRC/HELPERS

Reusable utility functions for working with multipart payloads, HTTP requests, and standardized responses.

src/helpers/multipart.ts

[HELPERS](#)

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>parseMultipart(event)</code>	<i>No doc comment</i> Params: <code>event</code> : APIGatewayProxyEventV2 Returns: <code>ParsedMultipart</code>

src/helpers/request.ts

[HELPERS](#)

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>parseBody(event)</code>	<i>No doc comment</i> Params: <code>event</code> : APIGatewayProxyEventV2 Returns: <code>Record</code>
export <code>parseEvent(event)</code>	<i>No doc comment</i> Params: <code>event</code> : APIGatewayProxyEventV2 Returns: <code>ParsedRequest</code>
export <code>getCookieValue(event, cookieName)</code>	<i>No doc comment</i> Params: <code>event</code> : APIGatewayProxyEventV2, <code>cookieName</code> : string Returns: <code>string</code> <code>undefined</code>
export <code>getAuthorizationToken(event)</code>	<i>No doc comment</i> Params: <code>event</code> : APIGatewayProxyEventV2 Returns: <code>string</code> <code>undefined</code>
export <code>getPaginationParams(queryStringParameters)</code>	<i>No doc comment</i> Params: <code>queryStringParameters</code> : <code>Record</code> Returns: <code>{ page: number; limit: number; offset: number }</code>

src/helpers/response.ts

[HELPERS](#)

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local getOrigin (eventOrOrigin?)	<i>No doc comment</i> Params: eventOrOrigin?: string APIGatewayProxyEventV2 Returns: string undefined
local buildHeaders (eventOrOrigin?)	<i>No doc comment</i> Params: eventOrOrigin?: string APIGatewayProxyEventV2 Returns: Record

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
response	any	<i>No description</i>

SRC/MIDDLEWARE

HTTP intermediate logic processing (e.g., authentication verify, schema validations, error catching).

src/middleware/auth.ts

MIDDLEWARE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export authenticate(event)</code>	No doc comment Params: event: APIGatewayProxyEventV2 Returns: Promise
<code>export requireRole(user, allowedRoles)</code>	No doc comment Params: user: AuthenticatedUser, allowedRoles: string[] Returns: void

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>authenticateUser</code>	any	No description

src/middleware/errorHandler.ts

MIDDLEWARE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export errorHandler(handler)</code>	No doc comment Params: handler: AsyncHandler Returns: Handler

src/middleware/validate.ts

MIDDLEWARE

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local formatZodErrors (zodError)	<i>No doc comment</i> Params: zodError: ZodError Returns: Array<{ field: string; message: string }>
export validate (schema, data)	<i>No doc comment</i> Params: schema: ZodSchema, data: unknown Returns: T

SRC/REPOSITORIES

Data Access Layer encapsulating direct database operations via Drizzle ORM.

src/repositories/admin.repository.ts

REPOSITORIES

No description provided in file comments.

class AdminRepository

— Admin Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public listCandidates(filter)	— 1. Paginated candidate list with filters — Params: filter: CandidateListFilter Returns: Promise
public getCandidateDetail(candidateId)	— 2. Get single candidate with full detail — Params: candidateId: string Returns: Promise
public getCandidateDocuments(candidateId, page, limit)	— 3. Get candidate documents (paginated) — Params: candidateId: string, page: number, limit: number Returns: Promise<{ data: any[]; total: number; page: number; limit: number; totalPages: number }>
public getStats()	— 4. Get statistics / summary — Params: none Returns: Promise
public listCandidatesForExport(filter)	— 5. Get all candidates for XLSX export (no pagination) — Params: filter: Omit Returns: Promise
public verifyDocument(documentId, isVerified)	— 6. Update document verification status — Params: documentId: string, isVerified: boolean Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
adminRepository	any	No description

src/repositories/application.repository.ts

REPOSITORIES

No description provided in file comments.

class ApplicationRepository

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public findByCandidateId (candidateId)	— Applications — Params: candidateId: string Returns: Promise
public findById (id)	No doc comment Params: id: string Returns: Promise
public findDraftByCandidateId (candidateId)	No doc comment Params: candidateId: string Returns: Promise
public create (data)	No doc comment Params: data: NewApplication Returns: Promise
public updateStatus (applicationId, status)	No doc comment Params: applicationId: string, status: string Returns: Promise
public updateCurrentStep (applicationId, currentStep, completedSteps)	No doc comment Params: applicationId: string, currentStep: number, completedSteps: number[] Returns: Promise
public submit (applicationId, referenceNumber)	No doc comment Params: applicationId: string, referenceNumber: string Returns: Promise
public getStepData (applicationId, stepNumber)	— Application Step Data — Params: applicationId: string, stepNumber: number Returns: Promise
public getAllStepData (applicationId)	No doc comment Params: applicationId: string Returns: Promise
public upsertStepData (applicationId, stepNumber, data)	No doc comment Params: applicationId: string, stepNumber: number, data: Record Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
applicationRepository	any	No description

src/repositories/common.repository.ts

REPOSITORIES

No description provided in file comments.

class `OtpRepository`

— OTP Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>create(data)</code>	<i>No doc comment</i> Params: <code>data: NewOtp</code> Returns: <code>Promise</code>
public <code>findById(otpRequestId)</code>	<i>No doc comment</i> Params: <code>otpRequestId: string</code> Returns: <code>Promise</code>
public <code>findByToken(token)</code>	<i>No doc comment</i> Params: <code>token: string</code> Returns: <code>Promise</code>
public <code>markVerified(otpId, token)</code>	<i>No doc comment</i> Params: <code>otpId: string, token: string</code> Returns: <code>Promise</code>
public <code>updateResendTime(otpId, newResendTime)</code>	<i>No doc comment</i> Params: <code>otpId: string, newResendTime: Date</code> Returns: <code>Promise</code>
public <code>deleteExpired()</code>	<i>No doc comment</i> Params: <code>none</code> Returns: <code>Promise</code>

class `CaptchaRepository`

— CAPTCHA Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>create(data)</code>	<i>No doc comment</i> Params: <code>data: NewCaptcha</code> Returns: <code>Promise</code>
public <code>findById(id)</code>	<i>No doc comment</i> Params: <code>id: string</code> Returns: <code>Promise</code>
public <code>delete(id)</code>	<i>No doc comment</i> Params: <code>id: string</code> Returns: <code>Promise</code>
public <code>deleteExpired()</code>	<i>No doc comment</i> Params: <code>none</code> Returns: <code>Promise</code>

class `DocumentRepository`

— Document Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public create (data)	<i>No doc comment</i> Params: data: NewDocument Returns: Promise
public findById (id)	<i>No doc comment</i> Params: id: string Returns: Promise
public findByCandidateId (candidateId)	<i>No doc comment</i> Params: candidateId: string Returns: Promise
public findByCandidateAndType (candidateId, documentType)	<i>No doc comment</i> Params: candidateId: string, documentType: string Returns: Promise
public delete (id)	<i>No doc comment</i> Params: id: string Returns: Promise
public update (id, data)	<i>No doc comment</i> Params: id: string, data: Partial Returns: Promise

class PaymentRepository

— Payment Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public create (data)	<i>No doc comment</i> Params: data: NewPayment Returns: Promise
public findById (id)	<i>No doc comment</i> Params: id: string Returns: Promise
public findByApplicationId (applicationId)	<i>No doc comment</i> Params: applicationId: string Returns: Promise
public findByOrderId (paymentOrderId)	<i>No doc comment</i> Params: paymentOrderId: string Returns: Promise
public findByRcptId (rcptId)	<i>No doc comment</i> Params: rcptId: string Returns: Promise
public updateStatus (paymentId, updates)	<i>No doc comment</i> Params: paymentId: string, updates: Partial< Pick > Returns: Promise

class InvoiceRepository

— Invoice Repository —

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public create(data)</code>	<i>No doc comment</i> Params: <code>data: NewInvoice</code> Returns: <code>Promise</code>
<code>public findById(id)</code>	<i>No doc comment</i> Params: <code>id: string</code> Returns: <code>Promise</code>
<code>public findByPaymentId(paymentId)</code>	<i>No doc comment</i> Params: <code>paymentId: string</code> Returns: <code>Promise</code>
<code>public findByInvoiceNumber(invoiceNumber)</code>	<i>No doc comment</i> Params: <code>invoiceNumber: string</code> Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>otpRepository</code>	any	— Singleton Instances
<code>captchaRepository</code>	any	<i>No description</i>
<code>documentRepository</code>	any	<i>No description</i>
<code>paymentRepository</code>	any	<i>No description</i>
<code>invoiceRepository</code>	any	<i>No description</i>

src/repositories/gatewayPayment.repository.ts

REPOSITORIES

No description provided in file comments.

`class GatewayPaymentRepository`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public findActiveGateway(gatewayName?)</code>	Gateways Params: <code>gatewayName?: string</code> Returns: <code>Promise</code>
<code>public createTransaction(data)</code>	Transactions Params: <code>data: NewPaymentTransaction</code> Returns: <code>Promise</code>
<code>public findTransactionById(transactionId)</code>	<i>No doc comment</i> Params: <code>transactionId: string</code> Returns: <code>Promise</code>

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public updateTransactionStatus (transactionId, status, gatewayResponse?)	<i>No doc comment</i> Params: transactionId: string, status: string, gatewayResponse?: Record Returns: Promise
public createLog (data)	Logs Params: data: NewPaymentLog Returns: Promise
public createRefund (data)	Refunds Params: data: NewPaymentRefund Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
gatewayPaymentRepository	any	No description

src/repositories/location.repository.ts

REPOSITORIES

No description provided in file comments.

class LocationRepository

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public findCountries (isActiveOnly)	— Countries — Params: isActiveOnly: any Returns: Promise
public findStatesByCountryId (countryId, isActiveOnly)	— States — Params: countryId: number, isActiveOnly: any Returns: Promise
public findStateByCode (stateCode, isActiveOnly)	<i>No doc comment</i> Params: stateCode: string, isActiveOnly: any Returns: Promise
public findDistrictsByStateId (stateId, isActiveOnly)	— Districts — Params: stateId: number, isActiveOnly: any Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
locationRepository	any	No description

src/repositories/user.repository.ts

REPOSITORIES

No description provided in file comments.

class UserRepository

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public findByEmail(email)	— Users Params: email: string Returns: Promise
public findById(id)	No doc comment Params: id: string Returns: Promise
public findByCognitoSubId(id)	No doc comment Params: id: string Returns: Promise
public create(data)	No doc comment Params: data: NewUser Returns: Promise
public updateLastLogin(userId)	No doc comment Params: userId: string Returns: Promise
public updatePassword(userId, passwordHash)	No doc comment Params: userId: string, passwordHash: string Returns: Promise
public deactivate(userId)	No doc comment Params: userId: string Returns: Promise
public findCandidateByUserId(userId)	— Candidates Params: userId: string Returns: Promise
public findCandidateById(id)	No doc comment Params: id: string Returns: Promise
public findCandidateByRegistrationNumber(regNo)	No doc comment Params: regNo: string Returns: Promise
public createCandidate(data)	No doc comment Params: data: NewCandidate Returns: Promise

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public updateCandidateVerification (candidateId, updates)	<i>No doc comment</i> Params: candidateId: string, updates: Partial< Returns: Promise
public updateCandidateDetails (candidateId, updates)	<i>No doc comment</i> Params: candidateId: string, updates: Partial< Returns: Promise
public updateCandidateRegistrationNumber (candidateId, registrationNumber)	<i>No doc comment</i> Params: candidateId: string, registrationNumber: string Returns: Promise
public findAllCandidates ()	<i>No doc comment</i> Params: none Returns: Promise
public findUserWithRole (userId)	<i>No doc comment</i> Params: userId: string Returns: Promise<(User & { roleName: string }) null>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
userRepository	any	<i>No description</i>

SRC/SCRIPTS

Submodule containing codebase files.

src/scripts/triggerMtsSms.ts

SCRIPTS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local triggerSms()	No doc comment Params: none Returns: any

SRC/SERVICES

Business Logic Layer coordinating transactional workflows, third-party adapters (Razorpay, GetEPay, AWS SES/SNS).

src/services/admin.service.ts

SERVICES

No description provided in file comments.

class AdminService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public listCandidates(filter)	— 1. List candidates with pagination + filters — Params: filter: CandidateListFilter Returns: any
public getCandidateDetail(candidateId)	— 2. Get full candidate detail — Params: candidateId: string Returns: any
public getCandidateDocuments(candidateId, page, limit)	— 3. List candidate documents (paginated) — Params: candidateId: string, page: number, limit: number Returns: any
public getStats()	— 4. Admin dashboard stats — Params: none Returns: any
public exportCandidatesXlsx(filter)	— 5. Export candidates to XLSX Buffer — Params: filter: Omit Returns: Promise
public verifyDocument(documentId, isVerified)	— 6. Verify / unverify a document — Params: documentId: string, isVerified: boolean Returns: any

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
adminService	any	No description

src/services/application.service.ts

SERVICES

No description provided in file comments.

class ApplicationService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public getOrCreateDraft (candidateId)	— Get or Create Draft — Params: candidateId: string Returns: Promise
private buildDraftResponse (application, stepDataMap)	No doc comment Params: application: Application, stepDataMap: Record< Returns: ApplicationDraft
public saveStep (applicationId, candidateId, stepNumber, data)	— Save Step — Params: applicationId: string, candidateId: string, stepNumber: number, data: Record Returns: Promise
public getStepData (applicationId, candidateId, stepNumber)	— Get Step Data — Params: applicationId: string, candidateId: string, stepNumber: number Returns: Promise null>
public submitApplication (applicationId, candidateId)	— Submit Application — Params: applicationId: string, candidateId: string Returns: Promise
public getApplicationStatus (applicationId, candidateId)	— Get Application Status — Params: applicationId: string, candidateId: string Returns: Promise
public getPrintPreview (applicationId, candidateId)	— Print Preview — Params: applicationId: string, candidateId: string Returns: Promise>
public getAllStepsData (candidateId, applicationId?)	No doc comment Params: candidateId: string, applicationId?: string Returns: Promise>
private enrichStepData (stepDataMap)	No doc comment Params: stepDataMap: Record< Returns: Promise
public finalSubmitLegacy (applicationId, candidateId)	No doc comment Params: applicationId: string, candidateId: string Returns: Promise
public unifiedFinalSubmit (applicationId, candidateId)	No doc comment Params: applicationId: string, candidateId: string Returns: Promise
public generateHtml (applicationId, candidateId)	No doc comment Params: applicationId: string, candidateId: string Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
applicationService	any	No description

src/services/auth.service.ts

SERVICES

No description provided in file comments.

class AuthService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public generateCaptcha()	— CAPTCHA — Params: none Returns: Promise<{ captchaId: string; svg: string }>
public validateCaptcha(captchaId, text)	No doc comment Params: captchaId: string, text: string Returns: Promise
public login(input)	— Login — Params: input: LoginInput Returns: Promise
public register(input)	— Register — Params: input: RegisterInput Returns: Promise
public candidateRegister(input)	No doc comment Params: input: CandidateRegisterInput Returns: Promise
public refreshToken(input)	— Refresh Token — Params: input: RefreshTokenInput Returns: Promise
public forgotPassword(input)	— Forgot Password — Params: input: ForgotPasswordInput Returns: Promise
public resetPassword(input)	— Reset Password — Params: input: ResetPasswordInput Returns: Promise
public changePassword(userId, input)	— Change Password — Params: userId: string, input: ChangePasswordInput Returns: Promise
public seedDummy()	— Seeding Dummy Data — Params: none Returns: Promise
public logout(accessToken)	— Logout — Params: accessToken: string Returns: Promise

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>authService</code>	any	No description

src/services/document.service.ts

SERVICES

No description provided in file comments.

class DocumentService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>generateUploadPresignedUrl(input)</code>	— Upload — Params: input: { candidateId: string; documentType: string; fileName: string; mimeType: string; fileSize: number; } Returns: any
public <code>uploadDocument(input)</code>	No doc comment Params: input: FileUploadInput Returns: Promise
public <code>generateUploadPresignedUrlStep5(input)</code>	No doc comment Params: input: { candidateId: string; userId: string; documentType: string; fileName: string; mimeType: string; fileSize: number; } Returns: any
public <code>uploadDocumentStep5(input)</code>	No doc comment Params: input: { candidateId: string; userId: string; documentType: string; fileName: string; mimeType: string; fileContent: Buffer; fileSize: number; } Returns: Promise
public <code>uploadGeneratedHtml(candidateId, fileContent, refNo)</code>	No doc comment Params: candidateId: string, fileContent: Buffer, refNo: string Returns: Promise
public <code>getDownloadUrl(documentId, candidateId)</code>	— Get Presigned URL — Params: documentId: string, candidateId: string Returns: Promise
public <code>getDocumentPresignedUrl(documentId, candidateId, mode)</code>	No doc comment Params: documentId: string, candidateId: string, mode: 'view' 'download' Returns: Promise<{ url: string; expiresIn: number }>
public <code>getPresignedUrl(fileUrl)</code>	No doc comment Params: fileUrl: string Returns: Promise
public <code>getCandidateDocuments(candidateId)</code>	— Get All Documents — Params: candidateId: string Returns: Promise

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public deleteDocument(documentId, candidateId)</code>	— Delete Params: <code>documentId: string</code>, <code>candidateId: string</code> Returns: <code>Promise</code>
<code>private deleteFromS3(fileUrl)</code>	No doc comment Params: <code>fileUrl: string</code> Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>documentService</code>	any	No description

src/services/gatewayPayment.service.ts

SERVICES

No description provided in file comments.

class GatewayPaymentService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public createPayment(input)</code>	No doc comment Params: <code>input: CreatePaymentInput</code> Returns: <code>any</code>
<code>public verifyPayment(input)</code>	No doc comment Params: <code>input: VerifyPaymentInput</code> Returns: <code>any</code>
<code>public getStatus(transactionId)</code>	No doc comment Params: <code>transactionId: string</code> Returns: <code>any</code>
<code>public processWebhook(gateway, payload, signature?)</code>	No doc comment Params: <code>gateway: string</code>, <code>payload: any</code>, <code>signature?: string</code> Returns: <code>any</code>
<code>public refundPayment(transactionId)</code>	No doc comment Params: <code>transactionId: string</code> Returns: <code>any</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>gatewayPaymentService</code>	any	No description

src/services/getepay.adapter.ts

SERVICES

No description provided in file comments.

class GetepayAdapter

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
private static getKeyAndIv()	No doc comment Params: none Returns: any
private static getAlgorithm(keyBuf)	No doc comment Params: keyBuf: Buffer Returns: any
public static encrypt(data, key, iv)	No doc comment Params: data: string, key: string, iv: string Returns: string
public static decrypt(data, key, iv)	No doc comment Params: data: string, key: string, iv: string Returns: string
public static createOrder(amount, transactionId, customerDetails)	No doc comment Params: amount: number, transactionId: string, customerDetails: { email: string; phone: string; name: string } Returns: any
public static verifyPayment(paymentId)	No doc comment Params: paymentId: string Returns: any

src/services/notification.service.ts

SERVICES

No description provided in file comments.

class NotificationService

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
private getSesClient()	No doc comment Params: none Returns: SESClient
private getSnsClient()	No doc comment Params: none Returns: SNSClient

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>renderMobileOtpSms(otp)</code>	— SMS Templates Params: <code>otp: string</code> Returns: <code>string</code>
public <code>renderEmailOtpSms(otp)</code>	No doc comment Params: <code>otp: string</code> Returns: <code>string</code>
public <code>renderSubmissionSuccessSms(details)</code>	No doc comment Params: <code>details: SubmissionSuccessSmsDetails</code> Returns: <code>string</code>
public <code>renderEmailOtpEmail(otp, examName?)</code>	— Email Templates Params: <code>otp: string, examName?: string</code> Returns: <code>{ subject: string; body: string }</code>
public <code>renderRegistrationSuccessEmail(details)</code>	No doc comment Params: <code>details: RegistrationSuccessDetails</code> Returns: <code>{ subject: string; body: string; }</code>
public <code>renderSubmissionSuccessEmail(details)</code>	— Email Templates Params: <code>details: SubmissionSuccessDetails</code> Returns: <code>{ subject: string; body: string; }</code>
public <code>sendSms(mobileNumber, message)</code>	— Dispatch Methods Params: <code>mobileNumber: string, message: string</code> Returns: <code>Promise</code>
public <code>sendEmail(email, subject, body)</code>	No doc comment Params: <code>email: string, subject: string, body: string</code> Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>notificationService</code>	any	No description

src/services/otp.service.ts

SERVICES

No description provided in file comments.

```
class OtpService
```

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>sendMobile0tp(input)</code>	— Send Mobile OTP — Params: <code>input: SendMobile0tpInput</code> Returns: <code>Promise</code>
public <code>verifyMobile0tp(input)</code>	— Verify Mobile OTP — Params: <code>input: VerifyMobile0tpInput</code> Returns: <code>Promise</code>
public <code>sendEmail0tp(input)</code>	— Send Email OTP — Params: <code>input: SendEmail0tpInput</code> Returns: <code>Promise</code>
public <code>verifyEmail0tp(input)</code>	— Verify Email OTP — Params: <code>input: VerifyEmail0tpInput</code> Returns: <code>Promise</code>
public <code>resend0tp(input)</code>	— Resend OTP — Params: <code>input: Resend0tpInput</code> Returns: <code>Promise</code>

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local <code>mockSendSms(mobile, otp)</code>	<i>No doc comment</i> Params: <code>mobile: string, otp: string</code> Returns: <code>Promise</code>
local <code>mockSendEmail(email, otp)</code>	<i>No doc comment</i> Params: <code>email: string, otp: string</code> Returns: <code>Promise</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
otpService	any	<i>No description</i>

src/services/payment.service.ts

SERVICES

No description provided in file comments.

class `PaymentService`

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
public <code>initiatePayment(input)</code>	— Initiate Payment — Params: <code>input: InitiatePaymentInput</code> Returns: <code>Promise</code>

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>public completeFreePayment(paymentOrderId)</code>	— Complete Free Payment — Params: <code>paymentOrderId: string</code> Returns: <code>Promise</code>
<code>public verifyPayment(input)</code>	— Verify Payment — Params: <code>input: VerifyPaymentInput</code> Returns: <code>Promise</code>
<code>public getPaymentStatus(applicationId, candidateId)</code>	— Get Payment Status — Params: <code>applicationId: string, candidateId: string</code> Returns: <code>Promise</code>
<code>public getFeeStructure()</code>	— Fee Structure — Params: <code>none</code> Returns: <code>Record</code>
<code>public getInvoice(paymentOrderId)</code>	— Get Invoice — Params: <code>paymentOrderId: string</code> Returns: <code>any</code>

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
<code>paymentService</code>	<code>any</code>	<i>No description</i>

src/services/razorpay.adapter.ts

SERVICES

No description provided in file comments.

class RazorpayAdapter

METHOD / SIGNATURE	DESCRIPTION & PARAMETERS
<code>private static getClient()</code>	<i>No doc comment</i> Params: <code>none</code> Returns: <code>any</code>
<code>public static createOrder(amount, currency, receipt)</code>	<i>No doc comment</i> Params: <code>amount: number, currency: string, receipt: string</code> Returns: <code>any</code>
<code>public static verifySignature(orderId, paymentId, signature)</code>	<i>No doc comment</i> Params: <code>orderId: string, paymentId: string, signature: string</code> Returns: <code>boolean</code>

SRC/TYPES

Common TypeScript definitions, models, and interface definitions.

src/types/index.ts

TYPES

```
=====
Shared Type Definitions for Candidate Portal Backend
=====
```

No classes, functions, or variables exported from this module.

SRC/UTILS

General utilities including age calculation, captcha generation, AWS Cognito helper, email sending, SMS integration, and PDF helper.

src/utils/age.ts

UTILS

Safely parses a date string (e.g., DD-MM-YYYY, YYYY-MM-DD, DD/MM/YYYY) into a Date object.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
<code>export parseBSSCDate(dateStr)</code>	Safely parses a date string (e.g., DD-MM-YYYY, YYYY-MM-DD, DD/MM/YYYY) into a Date object. Params: <code>dateStr: string Date</code> Returns: <code>Date</code>
<code>export calculateExactAge(dobInput)</code>	Calculates a candidate's age as of the BSSC reference date (August 1, 2025) or a provided reference date. Calculates a candidate's age exactly as of the BSSC reference date (August 1, 2025). @param <code>dobInput</code> Date of birth (string or Date) @returns Object containing years, months, and days Params: <code>dobInput: string Date</code> Returns: <code>{ years: number; months: number; days: number }</code>
<code>export calculateBSSCAge(dobInput, referenceDate?)</code>	Legacy: Calculates a candidate's age as of the BSSC reference date (August 1, 2025). @param <code>dobInput</code> Date of birth (string or Date) @param <code>referenceDate</code> Optional reference date to calculate age against @returns Age in years Params: <code>dobInput: string Date, referenceDate?: string Date</code> Returns: <code>number</code>
<code>export getBSSCAgeLimits(categoryValue, gender, isPwD, isExServiceman)</code>	Gets the valid age limits for a candidate based on their category, gender, and relaxations. Rules: - Minimum Age: 21 years (as of 01-08-2025) - Maximum Age: - Unreserved (UR) & EWS: 35 years - BC-I & BC-II (Male): 37 years - Female (UR/EWS/BC-I/BC-II): 38 years - Scheduled Caste (SC) & Scheduled Tribe (ST) (Male & Female): 40 years - Relaxations: - PwD candidates get a relaxation of 10 years in the maximum age limit. - Ex-servicemen get a relaxation of 5 years in the maximum age limit. Params: <code>categoryValue: string, gender: string, isPwD: boolean, isExServiceman: boolean</code> Returns: <code>AgeLimits</code>
<code>export checkBSSCEligibility(dobInput, minAge, maxAge, referenceDate?)</code>	Checks if a candidate is strictly within the age limits on the reference date. If overage by even 1 day, returns false. Params: <code>dobInput: string Date, minAge: number, maxAge: number, referenceDate?: string Date</code> Returns: <code>boolean</code>

src/utils/captcha.ts

UTILS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export generateCaptcha()	No doc comment Params: none Returns: CaptchaData
export verifyCaptchaText(userInput, storedText)	No doc comment Params: userInput: string, storedText: string Returns: boolean

src/utils/cognito.ts

[UTILS](#)

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
local computeSecretHash(username)	No doc comment Params: username: string Returns: string
local generateMockTokens(email, userId)	No doc comment Params: email: string, userId: string Returns: CognitoTokens
export cognitoLogin(email, password)	— Login Params: email: string, password: string Returns: Promise
export cognitoRegister(email, password, attributes)	— Register Params: email: string, password: string, attributes: Record Returns: Promise<{ userSub: string }>
export cognitoUpdateUserAttributes(cognitoSubId, attributes)	update using cognito sub id update password and mobile number Params: cognitoSubId: string, attributes: Record Returns: Promise
export cognitoRefreshToken(email, refreshToken)	— Refresh Token Params: email: string, refreshToken: string Returns: Promise
export cognitoForgotPassword(email)	— Forgot Password Params: email: string Returns: Promise

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export cognitoConfirmForgotPassword (email, confirmationCode, newPassword)	<i>No doc comment</i> Params: email: string, confirmationCode: string, newPassword: string Returns: Promise
export cognitoSignOut (accessToken)	— Sign Out Params: accessToken: string Returns: Promise
export cognitoConfirmSignUp (email, confirmationCode)	— Confirm Sign Up Params: email: string, confirmationCode: string Returns: Promise
export cognitoResendConfirmationCode (email)	— Resend Confirmation Code Params: email: string Returns: Promise
export cognitoAdminGetUser (email)	— Admin Operations Params: email: string Returns: Promise null>
export cognitoAdminDeleteUser (email)	<i>No doc comment</i> Params: email: string Returns: Promise
export cognitoAdminUpdateUserAttributes (email, attributes)	<i>No doc comment</i> Params: email: string, attributes: Record Returns: Promise
local getJwksClient (issuer)	<i>No doc comment</i> Params: issuer: string Returns: any
local getSigningKey (client, kid)	<i>No doc comment</i> Params: client: jwksRsa.JwksClient, kid: string Returns: Promise
export verifyJwt (token)	<i>No doc comment</i> Params: token: string Returns: Promise
export getUserBySub (subId)	<i>No doc comment</i> Params: subId: string Returns: Promise null>
export getUserByCognitoSubId (cognitoSub)	<i>No doc comment</i> Params: cognitoSub: string Returns: Promise null>
export cognitoAdminSetUserPassword (cognitoSubId, passwordStr)	<i>No doc comment</i> Params: cognitoSubId: string, passwordStr: string Returns: Promise

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export getCognitoSubByEmail(email)	No doc comment Params: email: string Returns: Promise
export getCognitoUserByEmail(email)	No doc comment Params: email: string Returns: Promise null>

src/utils/crypto.ts

UTILS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export generateHash(plainText)	No doc comment Params: plainText: string Returns: Promise
export compareHash(plainText, hash)	No doc comment Params: plainText: string, hash: string Returns: Promise
export generateOTP(length)	No doc comment Params: length: number Returns: string
export generateRandomToken()	No doc comment Params: none Returns: string
export generateUUID()	No doc comment Params: none Returns: string
export generateHmac(data, secret)	No doc comment Params: data: string, secret: string Returns: string
export generateRegistrationNumber(prefix)	No doc comment Params: prefix: string Returns: string

src/utils/email.ts

UTILS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>sendEmail(options)</code>	<i>No doc comment</i> Params: <code>options: SendEmailOptions</code> Returns: <code>Promise</code>
export <code>sendPaymentSuccessEmail(to, candidateName, examName, applicationNo, transactionId, amount)</code>	Sends the Successful Payment and Final Application Submission Email. Params: <code>to: string, candidateName: string, examName: string, applicationNo: string, transactionId: string, amount: number string</code> Returns: <code>Promise</code>

src/utils/pdf.ts

UTILS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>generateApplicationHtml(data)</code>	<i>No doc comment</i> Params: <code>data: PdfGenerationData</code> Returns: <code>Promise</code>

src/utils/sms.ts

UTILS

No description provided in file comments.

Functions

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export <code>sendSms(options)</code>	Sends an SMS using the AnalyticsMantra Bulk SMS gateway. @param options Contains the mobile number, message, and optional templateId. Params: <code>options: SmsOptions</code> Returns: <code>Promise</code>
export <code>sendMtsExamNotificationSms(mobile)</code>	Sends the specific MTS Exam Notification SMS to a given mobile number. Params: <code>mobile: string</code> Returns: <code>Promise</code>
export <code>sendMtsExamPortalOpenSms(mobile, fromDate, toDate)</code>	Sends the MTS Exam Portal Open Notification. Params: <code>mobile: string, fromDate: string, toDate: string</code> Returns: <code>Promise</code>

FUNCTION / SIGNATURE	DESCRIPTION & PARAMETERS
export sendPaymentSuccessSms (mobile, examName, amount, applicationNo)	Sends the Successful Payment SMS. Params: mobile: string, examName: string, amount: number string, applicationNo: string Returns: Promise

SRC/VALIDATORS

Zod verification schemas for request body and query parameter validation.

src/validators/application.ts

VALIDATORS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
step0Schema	any	No description
step1Schema	any	— Step 1: Reservation Category
step2Schema	any	— Step 2: Education
step3Schema	any	— Step 3: Post Preference
step4Schema	any	— Step 4: Language Selection
step5Schema	any	— Step 5: Examination Centre Selection
step6Schema	any	— Step 6: Document Upload
step7Schema	any	— Step 7: Fee Payment
step8Schema	any	— Step 8: Final Review & Submit
updateStepSchema	any	— Generic Step Update Schema
STEP_SCHEMAS	any	No description

src/validators/auth.ts

VALIDATORS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
loginSchema	any	— Auth Schemas
validateCaptchaSchema	any	No description
registerSchema	any	No description
candidateRegisterSchema	any	No description
forgotPasswordSchema	any	No description
resetPasswordSchema	any	No description
refreshTokenSchema	any	No description
changePasswordSchema	any	No description
sendMobile0tpSchema	any	— OTP Schemas
verifyMobile0tpSchema	any	No description
sendEmail0tpSchema	any	No description
verifyEmail0tpSchema	any	No description
resend0tpSchema	any	No description
candidateAddressSchema	any	— Candidate Step 1 Schema
candidateStep1Schema	any	No description
candidateStep2Schema	any	— Candidate Step 2 Schema
candidateStep3Schema	any	— Candidate Step 3 Schema
candidateStep4Schema	any	No description
candidateStep5Schema	any	No description
candidateStep6Schema	any	No description

src/validators/gatewayPayment.ts

VALIDATORS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
createPaymentSchema	any	No description

VARIABLE NAME	TYPE	DESCRIPTION
verifyPaymentSchema	any	No description
refundPaymentSchema	any	No description

src/validators/notification.ts

VALIDATORS

No description provided in file comments.

Exported Variables & Schemas

VARIABLE NAME	TYPE	DESCRIPTION
sendEmailNotificationSchema	any	No description